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Unit – 1 : Session – 11 : SLO – 1 | EX-OR gate

I. Match the Following

1. EX-OR Logic Expression — b) $A \oplus B = A \bar{B} + AB \bar{A}$
2. Output Condition — a) 0 when inputs are equal
3. Symbol Used — c) $\oplus$
4. Truth Table Output for (1,0) — d) 1
5. Major Application — e) Half Adder and Parity Checker

II. Fill in the Blanks

1. different
2. $A \bar{B} + AB \bar{A}$
3. $\oplus$
4. parity
5. EX-OR

III. Activity / Describe and Perform

- Logic operation: EX-OR outputs logic 1 only when the two inputs DIFFER. Boolean expression: $Y = A \oplus B = A \bar{B} + AB \bar{A}$.
- Truth table: $00 \rightarrow 0$, $01 \rightarrow 1$, $10 \rightarrow 1$, $11 \rightarrow 0$.
- Circuit symbol: OR symbol with an extra curved line at the input side.
- Practical application: parity generator/checker in data communication and the SUM output of a Half Adder (Sum = $A \oplus B$, Carry = $A \cdot B$). Experimental verification: applying all four input combinations to the gate reproduces the truth table exactly.